

# Assignment 2 Q1

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Q1

(a)

```
# Makes the rollset
set.seed(341)
rolls_fair = sample(1:6, size=12000, replace=TRUE)
rolls_unfair1 = sample(1:6, size=12000, replace=TRUE,
  prob=c(1,2,2,2,2,3)/12)
rolls_unfair2 = sample(1:6, size=12000, replace=TRUE,
  prob=c(3,4,4,4,4,5)/24)
rolls_unfair3 = sample(1:6, size=12000, replace=TRUE,
  prob=c(3,3,3,1,1,1)/12)

rho_ncp_fn = function(x=0, rollset) {

  observed <- table(factor(rollset, levels = 1:6))
  expected <- rep(12000 / 6, 6)
  chi_stat <- sum((observed - expected)^2 / expected)

  density_val <- dchisq(chi_stat, df = 5, ncp = x)

  return(-log(density_val))
}
```

(b)

```
fn1 <- function(x) rho_ncp_fn(x, rollset = rolls_fair)
fn2 <- function(x) rho_ncp_fn(x, rollset = rolls_unfair1)
fn3 <- function(x) rho_ncp_fn(x, rollset = rolls_unfair2)
fn4 <- function(x) rho_ncp_fn(x, rollset = rolls_unfair3)

result_fair = optim(par = 0, fn = fn1, method = "Brent", lower = 0, upper = 10)
result_unfair1 = optim(par = 0, fn = fn2, method = "Brent", lower = 0, upper = 4000)
result_unfair2 = optim(par = 0, fn = fn3, method = "Brent", lower = 0, upper = 4000)
```

```
result_unfair3 = optim(par = 0, fn = fn4, method = "Brent", lower = 0, upper = 4000)
result_fair$par
```

```
## [1] 0.9938273
```

```
result_unfair1$par
```

```
## [1] 1059.665
```

```
result_unfair2$par
```

```
## [1] 240.2617
```

```
result_unfair3$par
```

```
## [1] 3124.808
```

(c)

```
prob1=c(1,2,2,2,2,3)/12
prob2=c(3,4,4,4,4,5)/24
prob3=c(3,3,3,1,1,1)/12
fair_prob=rep(1/6, 6)

theta1=sum( (prob1 - fair_prob)^2 / fair_prob ) * 12000
theta2=sum( (prob2 - fair_prob)^2 / fair_prob ) * 12000
theta3=sum( (prob3 - fair_prob)^2 / fair_prob ) * 12000

print(theta1)
```

```
## [1] 1000
```

```
print(theta2)
```

```
## [1] 250
```

```
print(theta3)
```

```
## [1] 3000
```

These results show that the maximum likelihood estimates (MLEs) obtained via `optim()` align closely with the theoretical values derived from the known probability distributions. Small discrepancies between the MLEs and the true values are expected due to random sampling variation in the simulated data. In all three cases, however, the MLEs successfully reflect the magnitude and relative ordering of the true non-centrality parameters, indicating that the model appropriately captures the deviation from fairness.